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Analysis of Partially Metal-Clad and Dielectric Overlay-Loaded Diffused Optical Waveguide

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SUMMARY

The diffused optical waveguide is one of the most promising for use as an element of an integrated optical circuit, and optical functional devices such as an optical modulator and an optical switch can be composed of this waveguide. Analysis of this type of waveguide is difficult because the refractive index of the substrate is distributed continuously; there have been few studies on a waveguide with a metal cladding and a dielectric overlay and with the refractive index which is 2-dimensionally inhomogeneous. We propose an approximate scalar finite-element method which is applicable to a lossy diffused optical waveguide, and performed numerical analysis of a metal-clad and dielectric overlay-loaded diffused optical waveguide whose substrate has a 2-dimensionally inhomogeneous refractive index profile. The results are presented in this paper.

1. Introduction

Optical waveguides with various cross sections have been proposed as elements of the optical IC. The diffused optical waveguide is one of the most promising because it is easy to fabricate and it has a small propagation loss. This can be applied to optical functional devices such as modulators and switches using the electrooptic effect and the acoustooptic effect. Figure 1 shows a general structure of the diffused optical waveguide where the refractive index n(x, y) of the substrate is inhomogeneous in both the depth (y) and the lateral (x) directions. Here n_s is the refractive index of the substrate, Δn is its maximum variation caused by diffusion, and f(x, y) is a distribution function of the refractive index. To make use of the electrooptic effect, metal films with a complex refractive index n_m and thickness t are loaded a distance W from each other, sandwiching a dielectric overlay of refractive index n_f, thickness t, and width W. The propagation characteristics can be controlled through W. Here n_0 is a refractive index of the cover.

The effects of the metal cladding and the dielectric overlay on the propagation characteristics have been studied in terms of the effective index method [1 - 4] and the variational method [5], where, however, lateral variation of the refractive index is not taken into account. On the other hand, the effect of a 2-dimensionally inhomogeneous refractive index profile [6] is considered in the treatments via the variational method [7], the effective index method [8], the finite element method [9], and the linear segment layer approximation [10], in which, however, the metal cladding and the dielectric overlay are not considered.

The 2-dimensionally inhomogeneous distribution of the refractive index, the metal cladding, and the dielectric overlay all are taken into account in the present treatment to which an approximate finite-element analysis [11, 12] is applied. The vectorial finite-element method [9, 11, 13, 14] has been developed in terms of electric and magnetic fields along the propagation axis (z axis). However, it generates spurious modes that couple to the real modes [11, 14]. In addition, with a lossy medium, a number of computations must be repeated until an eigenvalue corresponding to the frequency becomes a real number. The approximate scalar finite-element metod, on the other hand, gives practically satisfactory results when lateral variation of the refractive index is small [15, 16], spurious modes are not generated, and it does not need repeated computations because the eigenvalue corresponds to the

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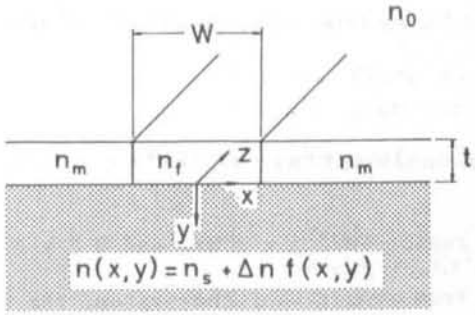


Fig. 1. Partially metal-clad and dielectric overlay-loaded diffused optical waveguide.

complex propagation constant itself. Therefore, the present method is easily applicable to a lossy diffused optical waveguide. Dispersion and attenuation characteristics of the waveguide as shown in Fig. 1 will be discussed numerically and the effect of the metal cladding and the dielectric overlay will be clarified. It is pointed out that the metal cladding does not influence greatly the dispersion characterstics, and the dispersion and the attenuation characteristics of the waveguide can be controlled through width W of the dielectric overlay.

2. Application of Approximate Scalar Finite-Element Method

The cross section of the waveguide shown in Fig. 1 is divided into triangular elements with six nodal points and it is assumed to be an isotropic medium. The E_{pq}^x mode with main electric field component E_x and magnetic field component H_y and the E_{pq}^y mode with main electric field component E_y and magnetic field component H_x are approximated as $E_y \equiv 0$ (a leading function is E_x) and $H_y \equiv 0$ (a leading function is H_y), respectively; this is called scalar approximation [17]. Further, we assume with regard to the relative dielectric constant ε_e in element e, that $\partial\varepsilon_e/\partial x = 0$ and $\partial(1/\varepsilon_e)/\partial x = 0$, and obtain

$$\left.\begin{aligned}&\left[\sum_e \int_{\Omega_e}\left(\xi_e k_0^2 \boldsymbol{N}_e \boldsymbol{N}_e^T + \frac{\gamma^2}{\eta_e}\boldsymbol{N}_e \boldsymbol{N}_e^T - \frac{1}{\eta_e}\frac{\partial \boldsymbol{N}_e}{\partial x}\frac{\partial \boldsymbol{N}_e^T}{\partial x}\right.\right.\\ &\left.\left.-\frac{1}{\eta_e}\frac{\partial \boldsymbol{N}_e}{\partial y}\frac{\partial \boldsymbol{N}_e^T}{\partial y}\right) dx\,dy\right]\boldsymbol{X}=\boldsymbol{0}\\ &\xi_e=\begin{cases}\varepsilon_e & E_{pq}^x \text{ mode}\\ 1 & E_{pq}^y \text{ mode}\end{cases},\quad \eta_e=\begin{cases}1 & E_{pq}^x \text{ mode}\\ \varepsilon_e & E_{pq}^y \text{ mode}\end{cases}\end{aligned}\right\}\quad(1)$$

where X represents a set of E_x at all nodal points in the E_{pq}^x mode or a set of H_x in the E_{pq}^y mode, p and q refer to the mode numbers in the x and y directions, respectively, N is the shape function, 0 is a zero vector, k_0 is the wave number in free space, $\gamma = \alpha + j\beta$, α is the attenuation constant, β is the phase constant in the axial direction, respectively, Ω_e dxdy denotes a surface integral over element e, $\sum_e$ denotes taking sum over all the elements in the cross section, and T specifies vector transpose.

Equation (1) can be solved as a complex eigenvalue problem with eigenvalue $(\gamma/k_0)^2$ and eigenvector X. This equation agrees with the exact matrix equation for a planar waveguide if we put $\partial/\partial x = 0$ [18]. Since the dielectric constant is distributed continuously in each element, numerical integration must be made or the constant dielectric constant must be assumed in the element in evaluating the matrix elements in Eq. (1); in either case, it takes computation time and requires a large computer memory. For the sake of efficiency, we approximate as

$$\left.\begin{aligned}\varepsilon_e &= \sum_{i=1}^{6} N_i\,\varepsilon_i && E_{pq}^x \text{ mode}\\ \varepsilon_e^{-1} &= \sum_{i=1}^{6} N_i\,\varepsilon_i^{-1} && E_{pq}^y \text{ mode}\end{aligned}\right\}\quad(2)$$

This approximation is better in accuracy and in computing time than the approximation in which the dielectric constant of each element is replaced by its mean value [9].

3. Analysis of a Diffused Optical Waveguide

3.1 Planar diffused optical waveguide

To check the validity of the present method, we apply the present method to a planar optical waveguide with an exponential index profile because we know its exact solution; in this case, in Fig. 1 we put $n_m = n_f = n_0$ and $f(x, y) = \exp(-y/d)$ where d is the diffusion length, $n_0 = 1.0$, $n_s = 2.2$, and $\Delta n = 0.01$. Figure 2 shows the element division and Fig. 3 shows the result for the TE modes; the fundamental mode (TE_0) and the higher-order modes (TE_1 and TE_2) agree with the exact solutions [19] up the cutoff frequency, and this agreement shows the validity of the present method.

3.2 Diffused optical waveguide

The solution to propagation characteristics of the diffused optical waveguide whose refractive index has a 2-dimensional profile has been obtained by means of the vectorial finite-element method when it has no metal cladding and no dielectric overlay, namely, $n_m = n_f = n_0 = 1.0$. Here we have [9]

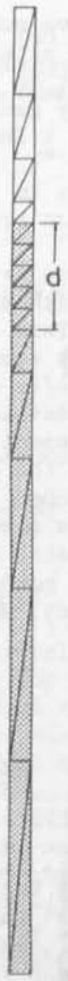


Fig. 2. Element division for planar diffused optical waveguide.

$$n(x, y) = n_2 + (n_2 - m_{1m})(y^2 - d^2)/d^2 \tag{3}$$

or

$$n(x, y) = n_2 + (n_2 - n_{1m})(x^2 + y^2 - l^2)/l^2 \tag{4}$$

for a region $-W/2 \leqq x \leqq W/2$ and $0 \leqq y \leqq d$.

From symmetry of the system, the section for $x \geqq 0$ is divided as in Fig. 4. The result of the analysis for the E_{11}^x and E_{11}^y modes is shown in Fig. 5; convergence of the result

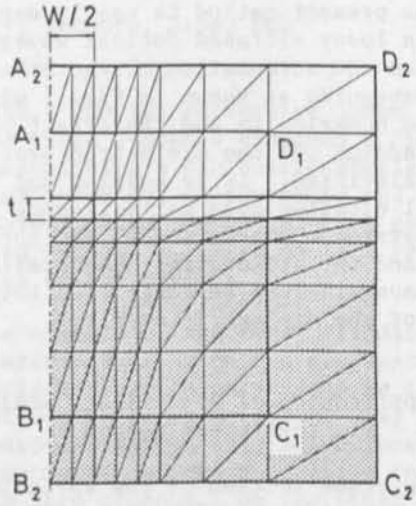


Fig. 4. Element division for the system of Fig. 1.

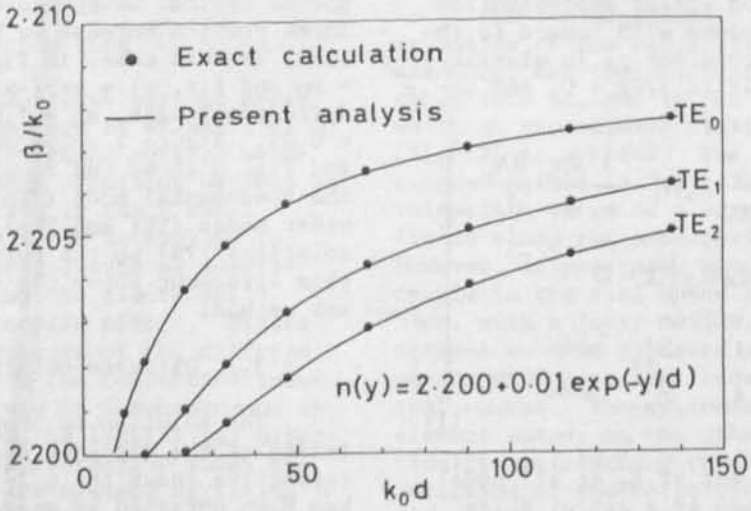


Fig. 3. Dispersion characteristics for planar diffused optical waveguide.

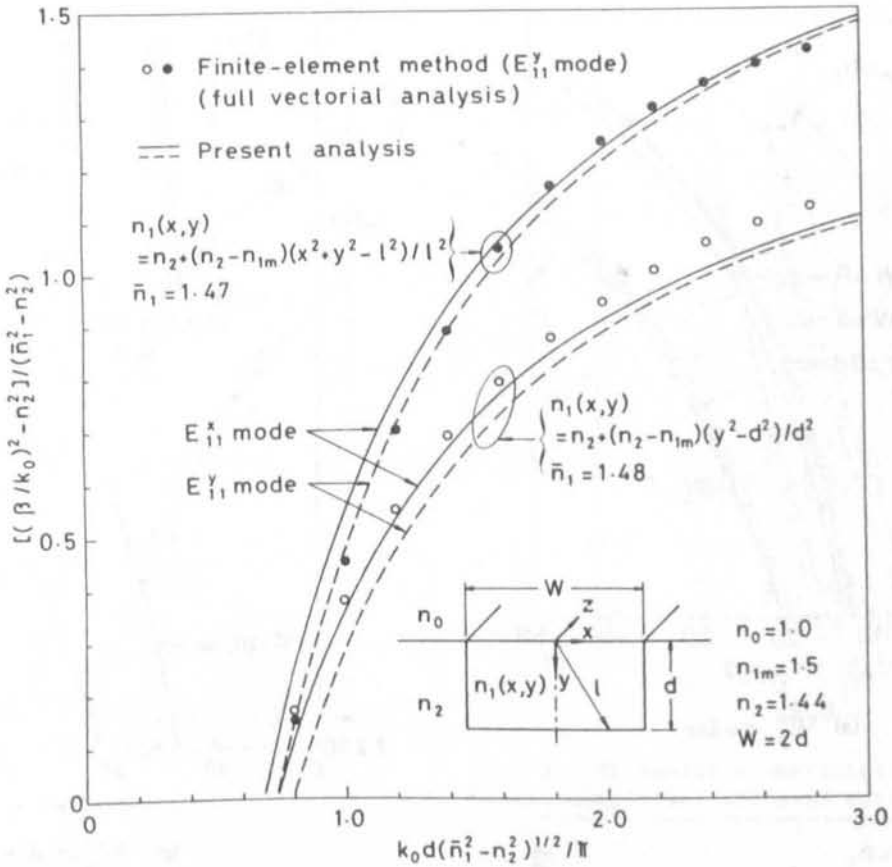


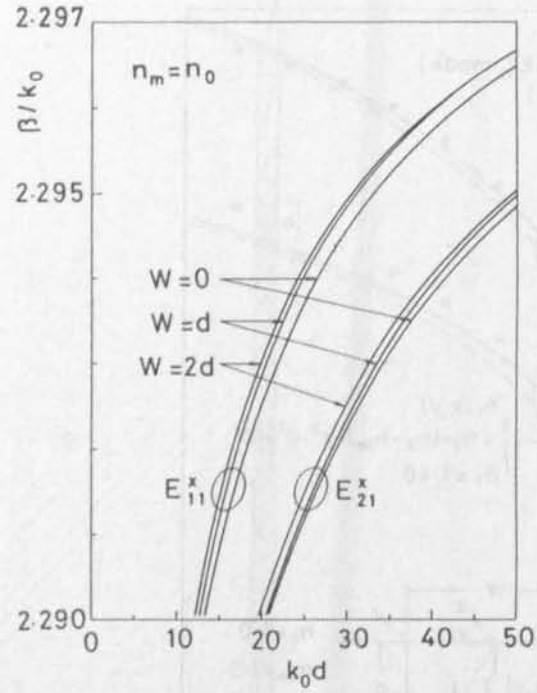
Fig. 5. Dispersion characteristics for diffused optical waveguide.

has been confirmed by extending the section $A_1B_1C_1D_1$ in Fig. 4 to $A_2B_2C_2D_2$. When the present result is compared with the result of the vectorial finite-element method made by Yeh et al. [9], good agreement is observed with a profile given by Eq. (4). However, some discrepancy is seen with a profile given by Eq. (3), probably for the following reasons. The refractive index in Eq. (4) varies continuously in the lateral direction, while in Eq. (3) it varies discontinuously at $x = \pm W/2$. In other words, the lateral variation of the refractive index in Eq. (3) is larger than that in Eq. (4), so that the validity of the approximate scalar finite-element method is poorer. In practice, the refractive index varies continuously, and the present method seems to be a good approximation to the pratical situation. For higher frequencies, the field energy is more concentrated in the central part and hence more elements are needed here. A discrepancy at higher frequencies between the present result on the index profile of Eq. (3) and the result of the vectorial finite-element analysis is attributable to the number of elements of the cross section.

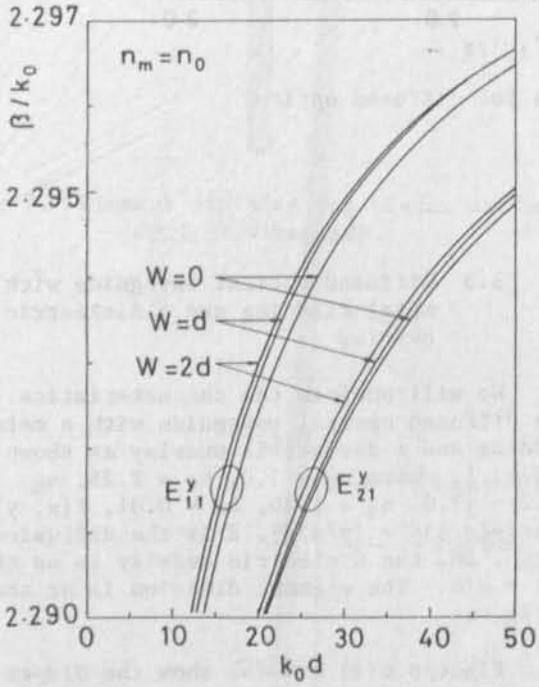
3.3 Diffused optical waveguide with a metal cladding and a dielectric overlay

We will analyze the characteristics of a diffused optical waveguide with a metal cladding and a dielectric overlay as shown in Fig. 1, where $n_0 = 1.0$, $n_f = 2.25$, $n_m = 1.2 - j7.0$, $n_s = 2.29$, $\Delta n = 0.01$, $f(x, y) = \exp(-(x/d)^2 - (y/d)^2)$, d is the diffusion length, and the dielectric overlay is as thick as $t = d/4$. The element division is as shown in Fig. 4.

Figures 6(a) and (b) show the dispersion characteristics of the $E_{pq}{}^x$ and $E_{pq}{}^y$ modes in a partially dielectric overlay-loaded diffused optical waveguide, where β/k_0 becomes larger with increase in W. The bandwidth for single-mode transmission is not affected greatly by W; this property permits some allowance in fabrication accuracy of the waveguide. Figures 7(a) and (b) show the dispersion characteristics of the $E_{pq}{}^x$ and $E_{pq}{}^y$ modes or when a metal cladding only is loaded; β/k_0 is decreased with decrease in W because decrease in W while keeping d and n

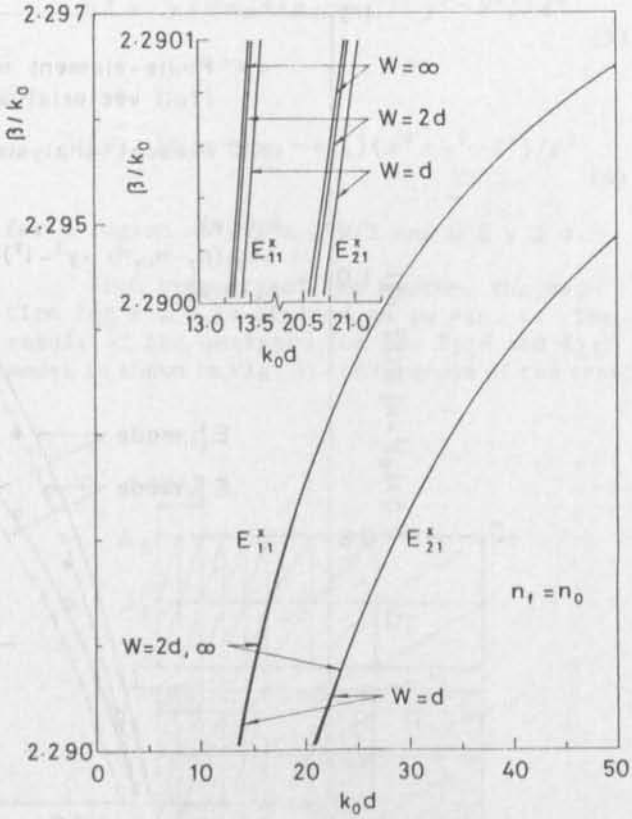


(a) E^x_{pq} modes.

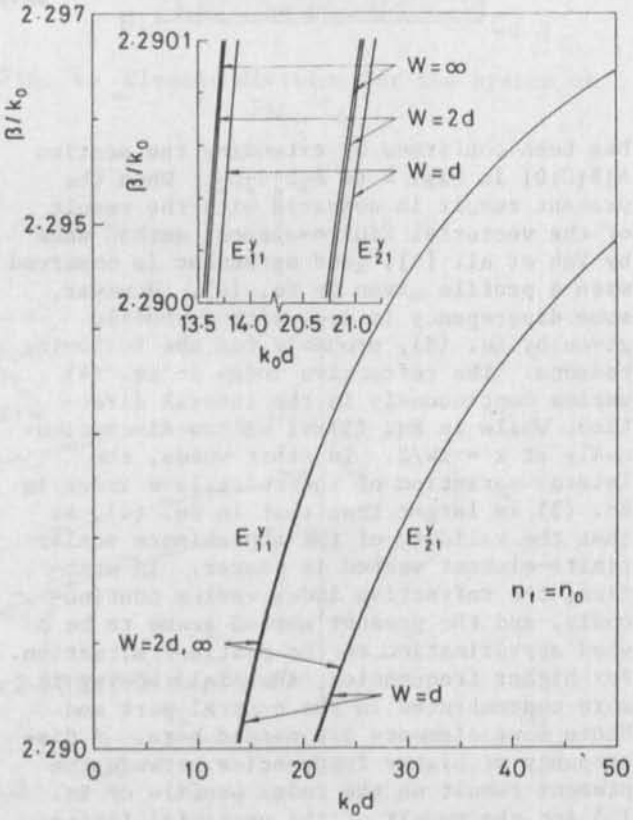


(b) E^y_{pq} modes.

Fig. 6. Dispersion characteristics for partially dielectric overlay-loaded diffused optical waveguide.

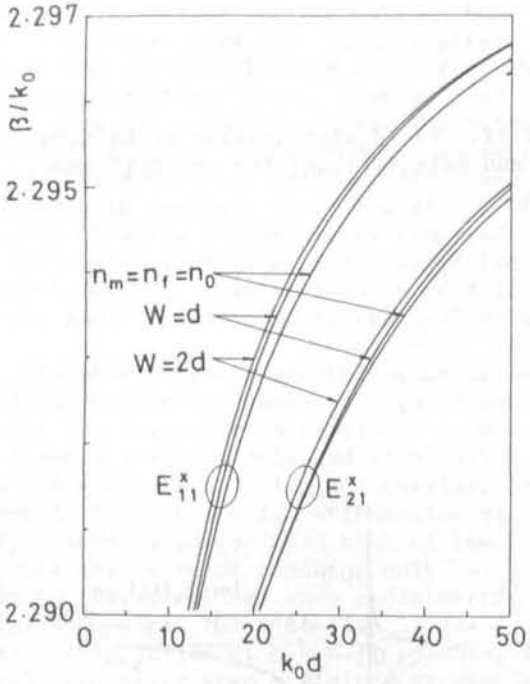


(a) E^x_{pq} modes.

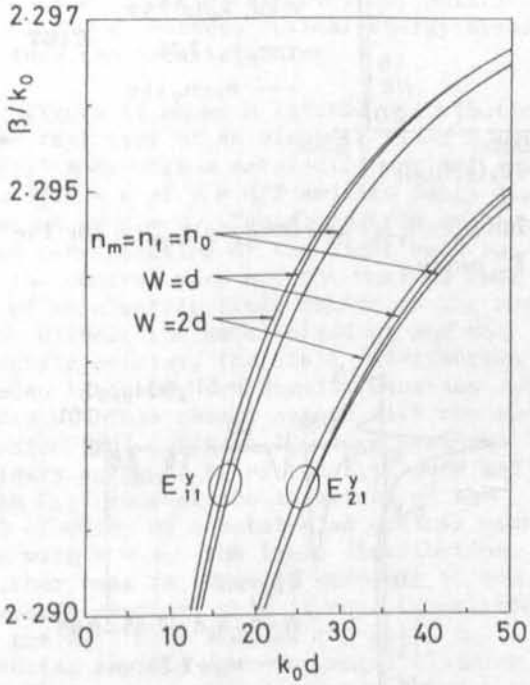


(b) E^y_{pq} modes.

Fig. 7. Dispersion characteristics for partially metal-clad diffused optical waveguide.



(a) E^x_{pq} modes.



(b) E^y_{pq} modes.

Fig. 8. Dispersion characteristics for partially metal-clad and dielectric overlay-loaded diffused optical waveguide.

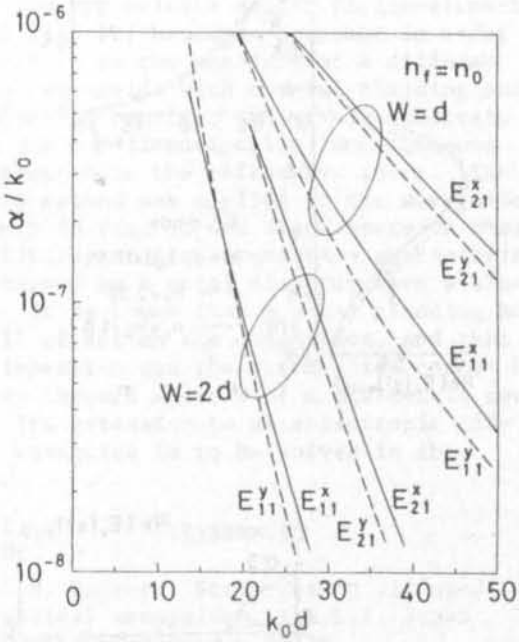


Fig. 9. Attenuation characteristics for partially metal-clad diffused optical waveguide.

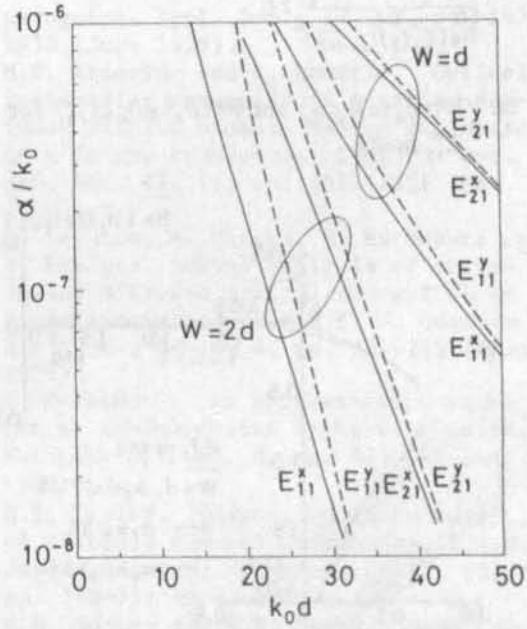


Fig. 10. Attenuation characteristics for partially metal-clad and dielectric overlay-loaded diffused optical waveguide.

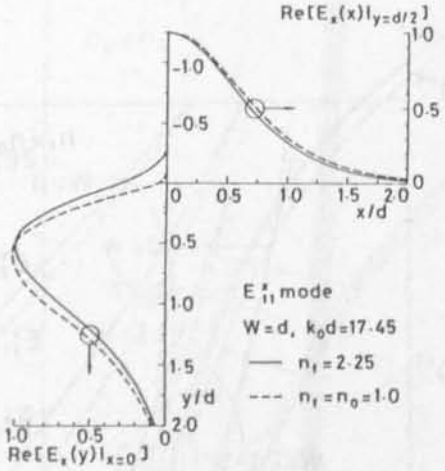
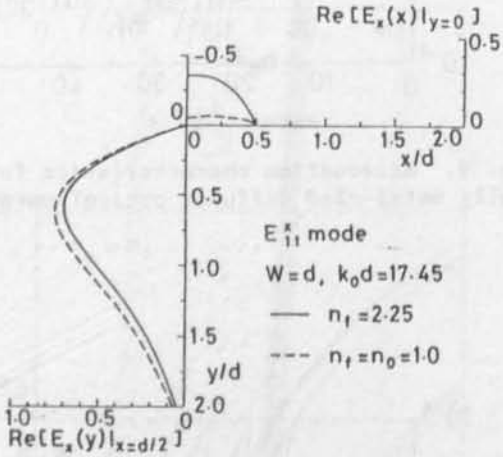
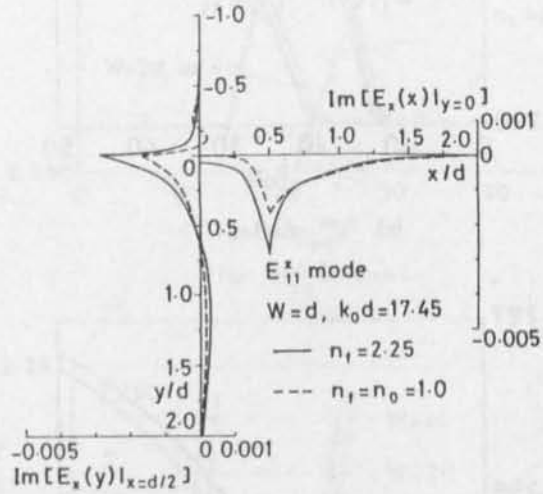


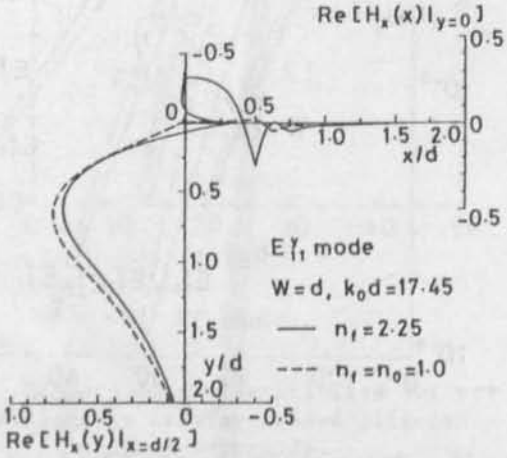
Fig. 11. Field distribution of $Re[E_x(x)|_{y=d/2}]$ and $Re[E_x(y)|_{x=0}]$ for the E_{11}^x mode.



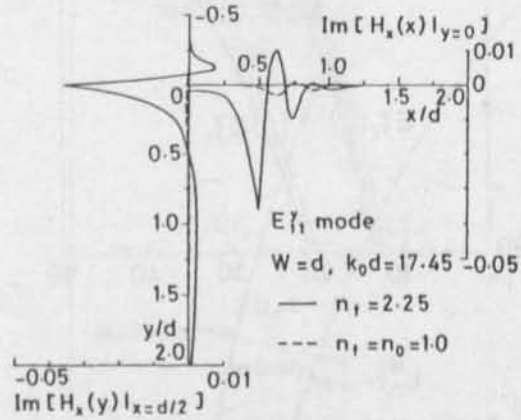
(a) $Re[E_x(x)|_{y=0}]$ and $Re[E_x(y)|_{x=d/2}]$ for the E_{11}^x mode.



(b) $Im[E_x(x)|_{y=0}]$ and $Im[E_x(y)|_{x=d/2}]$ for the E_{11}^x mode.



(c) $Re[H_x(x)|_{y=0}]$ and $Re[H_x(y)|_{x=d/2}]$ for the E_{11}^y mode.



(d) $Im[H_x(x)|_{y=0}]$ and $Im[H_x(y)|_{x=d/2}]$ for the E_{11}^y mode.

Fig. 12. Field distributions.

constant, enhances leakage of the light beam into the metal cladding and hence the effect of the metal cladding [2] increases. However, as is clear from Fig. 7, this effect is small and essentially does not affect the dispersion characteristics. Figures 8(a) and (b) show the dispersion characteristics of the E_{pq}^x and E_{pq}^y modes when the waveguide is loaded with both the metal cladding and the dielectric overlay, where the effect of the metal cladding is small when compared with the result in Fig. 6. The cutoff frequency of the E_{pq}^x mode in Figs. 6 to 8 is slightly smaller than that of the E_{pq}^y mode.

The attenuation characteristics of the optical waveguide are shown in Figs. 9 and 10 where the waveguide is partially loaded with a metal cladding only and with both a metal cladding and a dielectric overlay. As is seen in Figs. 9 and 10, attenuation of the E_{pq}^x mode is larger than that of the E_{pq}^y mode when a metal cladding only is loaded on the waveguide; when a dielectric overlay is loaded, the situation is reversed. Attenuation of a specified mode, in general, is larger when a dielectric overlay is loaded regardless of the value of W. Attenuation becomes larger as the lateral mode number p is increased independently of the existence of a dielectric overlay, polarization, and W, because optical energy spreads more into the metal cladding.

Figure 11 shows a lateral distribution of the real part of an electric field E_x of the E_{11}^x mode with a metal-clad optical waveguide of W = d at y = d/2 and its depth distribution at x = 0. The dielectric overlay causes concentration of the light beam toward the central part and the maximum real part of an electric field shifts to the surface. Without the metal cladding and the dielectric overlay, the field distribution is Gaussian laterally and Hermite-Gaussian into the depth. This result agrees with the observation [20]. Figure 12 shows real and imaginary parts of E_x of the E_{11}^x mode and H_x of the E_{11}^y mode at the interface of the metal cladding of a metal-clad optical waveguide with W = d. The field distribution in either case is affected strongly by the dielectric overlay; this is more remarkable with the E_{11}^y mode because the field E_x is attenuated rapidly near the metal cladding, while H_x is concentrated there and penetrates deeply into the metal cladding. The depth distribution of H_x in the waveguide with the dielectric overlay resembles that in a metal-clad uniform slab optical waveguide [21], because by loading it, the field is concentrated near the surface and the field distribution becomes similar to that in a uniform slab waveguide.

4. Conclusions

The approximate scalar finite-element method [11, 12] has been proposed in order to apply it to the analysis of a diffused optical waveguide with a metal cladding and a dielectric overlay, and with a substrate which has a 2-dimensionally inhomogeneous distribution of the refractive index. The present method was applied to the waveguide as shown in Fig. 1, and its dispersion characteristics and its attenuation characteristics caused by a metal cladding were evaluated. It is found that a metal cladding has a small effect on the dispersion, and that the dispersion and the attenuation can be controlled through width W of a dielectric overlay. Its extension to an anisotropic diffused waveguide is to be solved in the future.

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